

Quantifying Routing Overhead in Quantum Circuit Compilation

A controlled comparison of hardware topologies — and of the labelling hiding inside one

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How much does restricted qubit connectivity inflate compiled quantum circuits, which interaction patterns are most sensitive to hardware topology — and how much of the measured effect is connectivity rather than how the physical qubits happen to be numbered?

A compiler study, not a physics demonstration. It measures compiler output under modeled connectivity constraints; it does not measure QPU fidelity and makes no quantum-advantage claim. **1,080 canonical compilations** plus 20,040 in a label-permutation control, all under Qiskit 2.5.1. Four circuit families × six logical sizes (4–24) × three topologies at 27 physical qubits × three optimization levels × five fixed seeds, with circuit, basis, level and seed held fixed while connectivity varies.

2.691× / 2.909×
Pooled median two-qubit depth penalty, line / heavy-hex

2.140× / 2.304×
The same, excluding the level-0 stratum the control rules out

1.000×
GHZ chain on the line, and circular EfficientSU2 on heavy-hex at n = 12 and 20

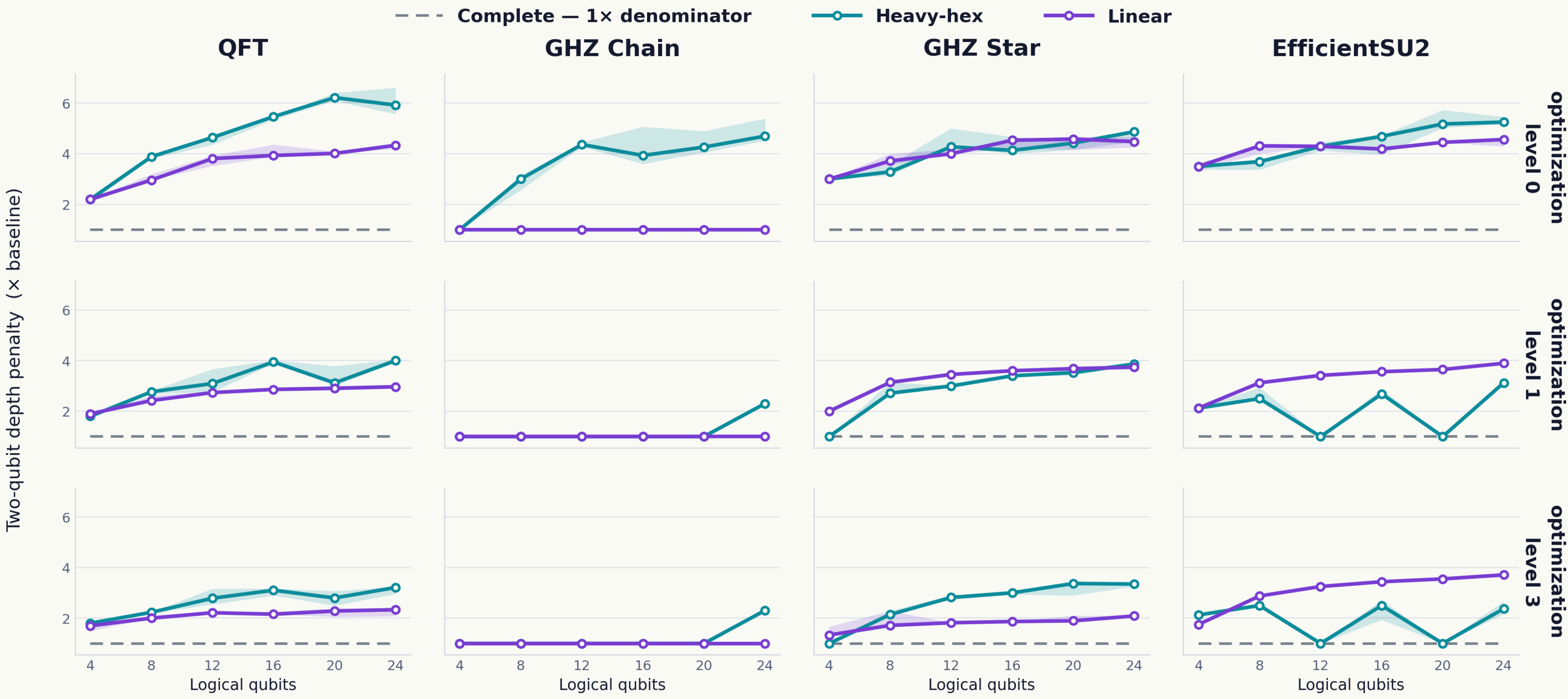
21 / {12, 20}
Heavy-hex longest simple path and cycle spectrum — forced by the graph

01 Primary result — two-qubit depth penalty

penalty = constrained-topology value ÷ matched complete-connectivity value · point = median of the 5 fixed seeds, band = their IQR · matching fixes family, size, circuit hash, level, seed and basis

Connectivity
Constrained connectivity commonly raises two-qubit depth: median **2.691×** on the line and **2.909×** on heavy-hex, 360 compilations each. Excluding level 0, **2.140×** and **2.304×**.

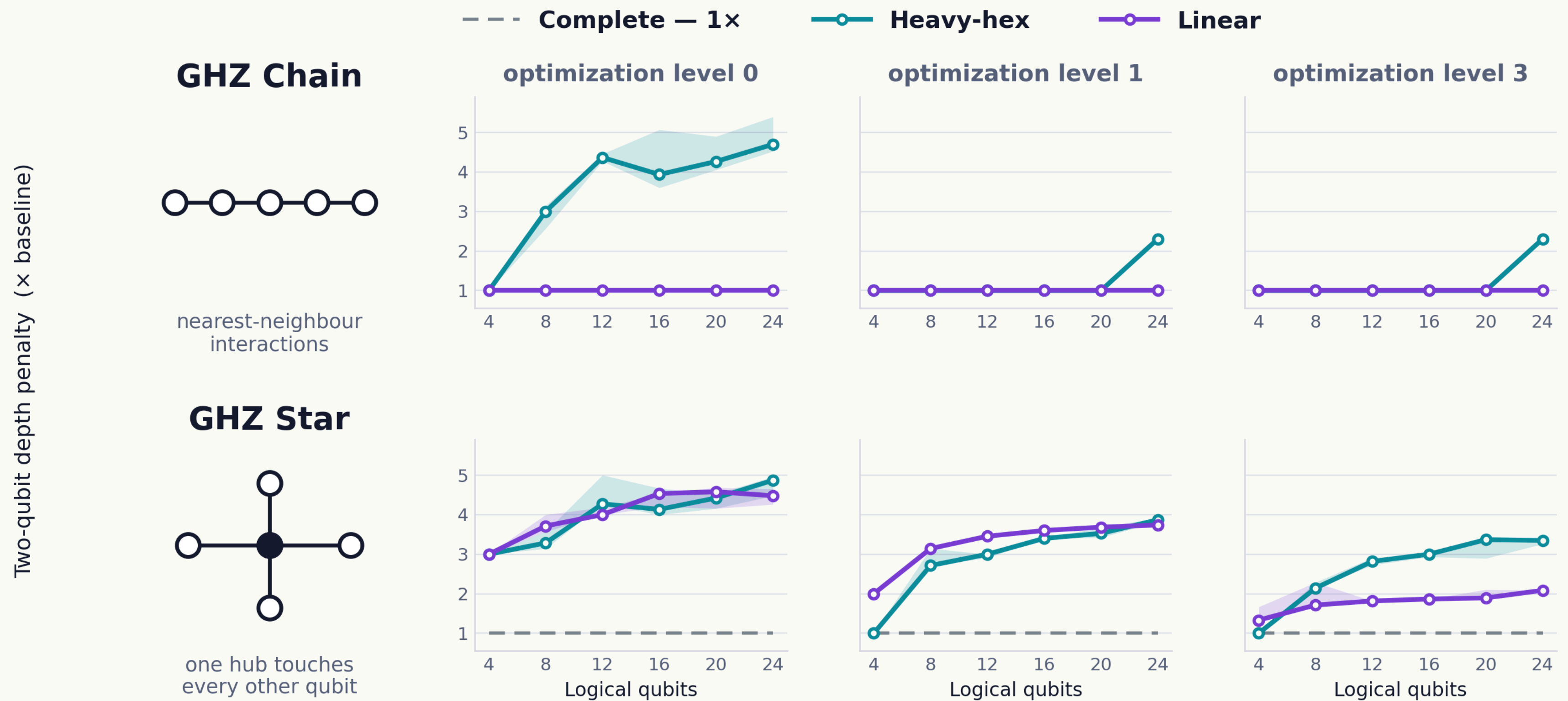
Optimization
Level 3 roughly halves the penalty against level 0 — **3.904×** → **1.991×** on the line, **4.282×** → **2.174×** on heavy-hex — at about 5× the local compile time.



02 Interaction structure matters

same logical task, same qubit count, same compiler settings — only the shape of the required interactions differs

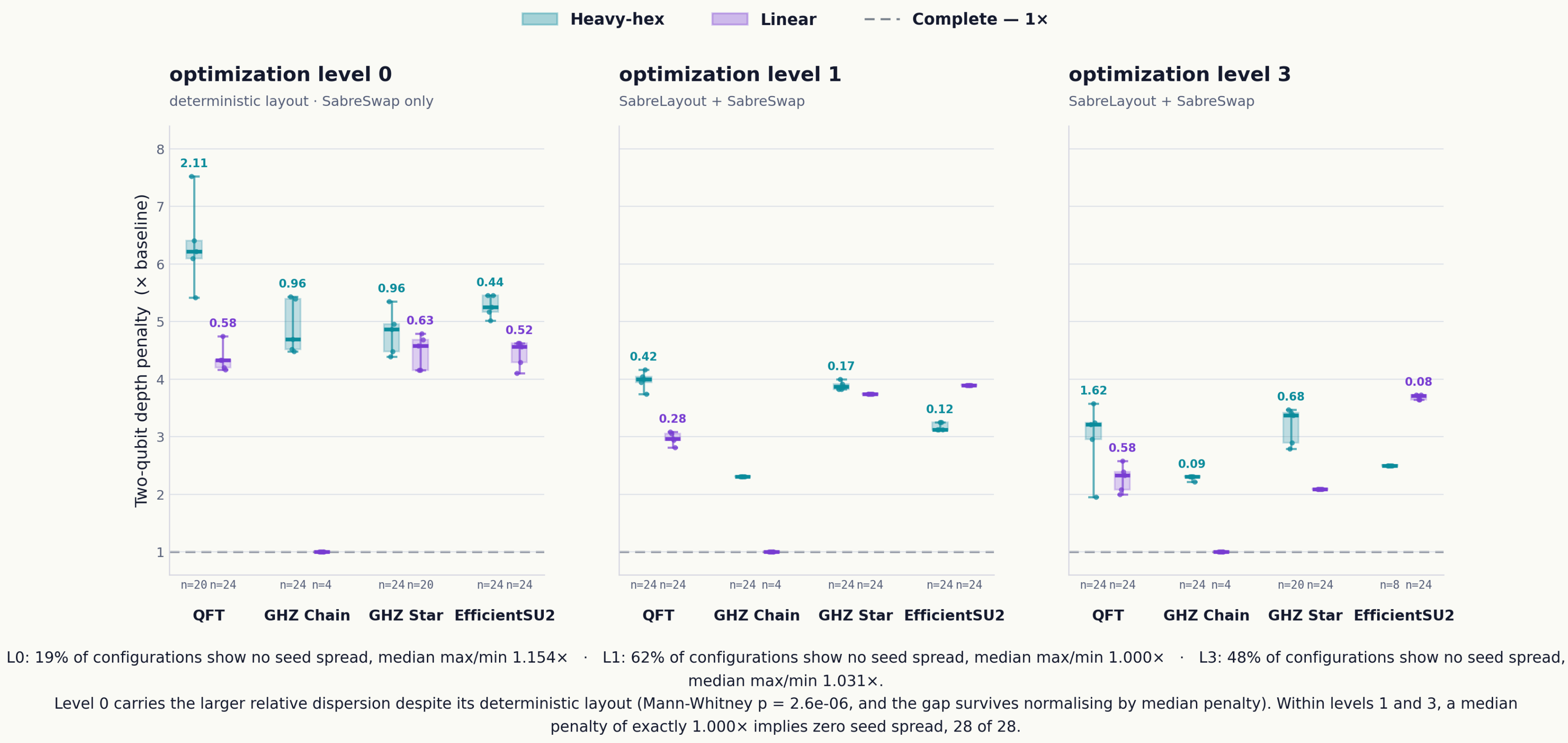
Circuit structure
Interaction shape dominates. GHZ chain on the line stays at exactly **1.000×** at every size and level, while GHZ star reaches **3.299×** on the same hardware — and the ranking reverses between QFT and EfficientSU2.



03 Seed variability, stratified by stochastic source

level 0 has a deterministic layout, so its spread is routing alone; levels 1 and 3 add stochastic SabreLayout · worst configuration per family and map, chosen independently within each level

Routing variability
Five seeds on QFT / heavy-hex / n = 20 / L0 span **5.419×** to **7.527×**. Level 0 carries the larger relative dispersion despite its deterministic layout (p = 2.6×10⁻⁶). A single-seed benchmark would have reported any point in that range.



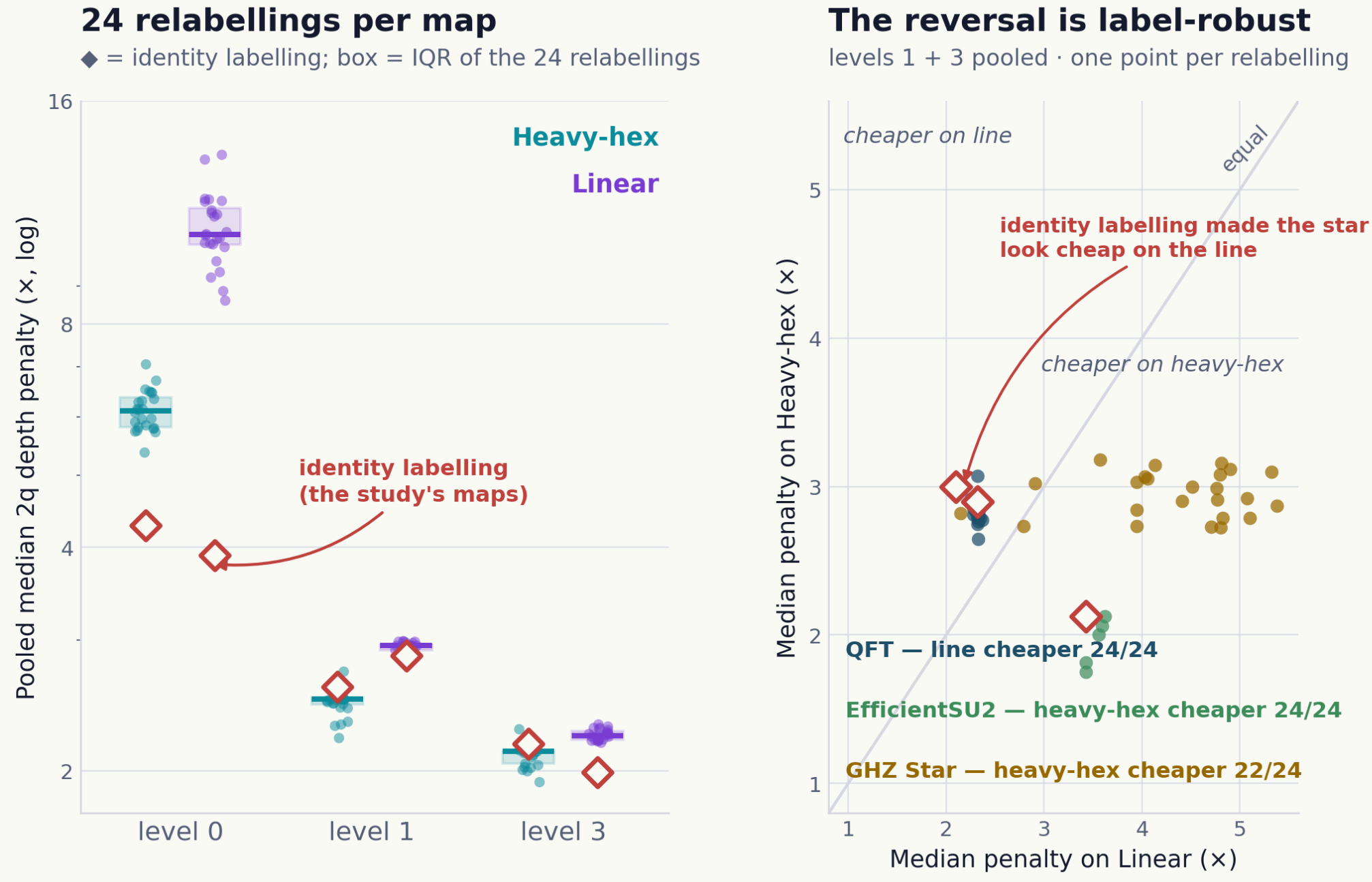
L0: 19% of configurations show no seed spread, median max/min 1.154× · L1: 62% of configurations show no seed spread, median max/min 1.000× · L3: 48% of configurations show no seed spread, median max/min 1.031×.
Level 0 carries the larger relative dispersion despite its deterministic layout (Mann-Whitney p = 2.6e-06, and the gap survives normalising by median penalty). Within levels 1 and 3, a median penalty of exactly 1.000× implies zero seed spread, 28 of 28.

04 The label-permutation control

24 deterministically derived relabellings per constrained map — same graph every time, only the numbering changes · SHA-256 derivation, reproducible on any platform · 17,280 compilations

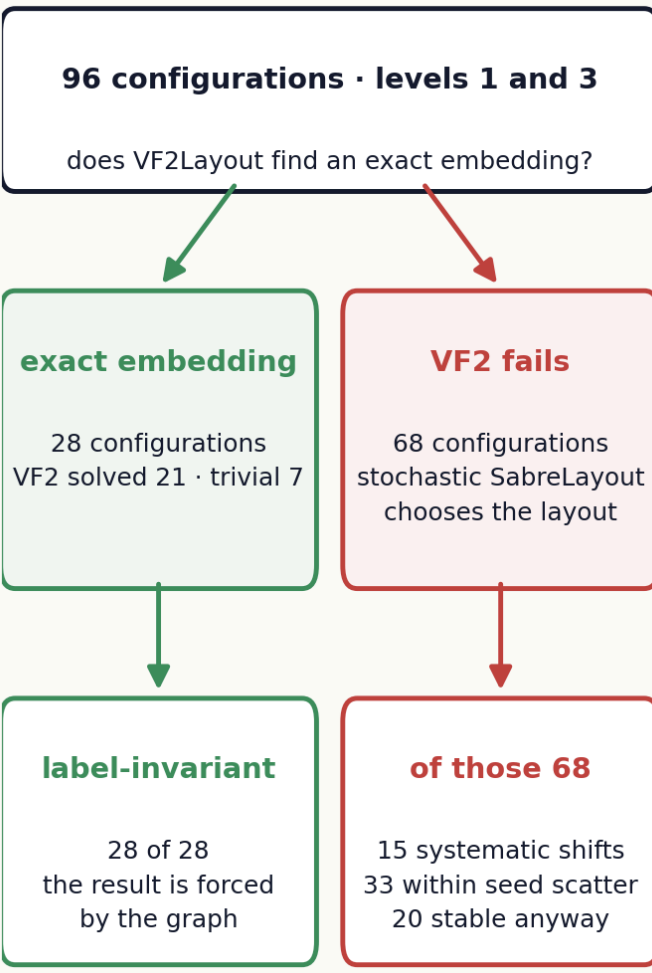
Labelling is a confound, and it is measurable
Relabelling the same graph moves level-0 results by up to **15.70×**; this study's labelling is more favourable than every relabelling tested. At levels 1 and 3 the spread falls to **1.06–1.23×** and the QFT / EfficientSU2 ranking holds in **24 of 24**. That number is reported as labelling, not connectivity.

And one claim did not survive
GHZ star looked cheap on the line under this study's labelling — **1.87×** at level 3, against a relabelling median of **3.99×**. Under relabelling the star is cheaper on heavy-hex in **22 of 24**. That number is reported as labelling, not connectivity.



Level 0: the identity labelling is more favourable than every relabelling tested, on both maps (relabelling medians 10.57× on line, 6.12× on heavy-hex). Levels 1 and 3: across-labelling spread falls to 1.06–1.23×, and the QFT / EfficientSU2 topology ranking holds in 24 of 24 relabellings.
GHZ star on the line at level 3 is the exception: identity gives 1.87× against a relabelling median of 3.99× (range 2.00–4.84×).

Why: the layout pass decides
instrumented VF2Layout_stop_reason, 960 compilations



every systematic shift falls in the right-hand branch · 0 exceptions

Limitations

- Compiler-level study, not physical hardware execution.
- One graph per topology class — heavy-hex results are Cairo's specific 27-node graph.
- Depends on the tested families, sizes, and the optimization set {0, 1, 3}.
- Compilation runtime is machine- and process-specific.
- Five seeds and 24 relabellings characterise, but do not exhaust, either space.

Methodological integrity

Level 0 measures labelling, not connectivity. Relabelling the same graph moves it by up to 15.70× — about three times the largest topology effect measured anywhere here — so no level-0 number carries a topology claim. Where an exact embedding exists the result is label-invariant, 28 of 28; every systematic shift falls where VF2Layout fails, 15 of 15. No exceptions in either direction.

Reproducible benchmark

Python 3.12 · Qiskit 2.5.1 · fixed seeds 11, 22, 33, 44, 55 · locked dependencies. Recompiling all 1,080 canonical points reproduced the stored metrics exactly, 1,080 of 1,080. Sweep conditions derive from a SHA-256 keystream, pinned by digest in the test suite.
github.com/auyjos/quantum-routing-overhead

